Kazakh-British Technical University
School of Information Technology and Engineering

SCADA systems and Industrial networks

SCADA CONTROLLERS AND THEIR INTEGRATIONS IN THE EXPERION PKS

Laboratory work №14

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CONTENT

Introduction..... 3
1. Tasks..... 4
1.1 Task 1. Installing SCADA controllers..... 4
1.2 Task 2. SCADA controller Status in the Station application..... 7
Conclusion..... 11
Worksheet..... 12

INTRODUCTION

The 14th lab shows how to integrate SCADA control units into the Experion Process Knowledge System (PKS) distributed control system. The theoretical part deals with the basics of programmable logic controllers (PLCs), which are compact electronic devices especially made to offer automated process control in the industrial sector. There is a special focus on the working benefits of PLCs, for example, their ability to run without human intervention for a very long time (up to 20 years), and their higher level of resistance even when exposed to extremely harsh environmental conditions which could be as low as -40C or as high as +70C. Besides that, the modular hardware architecture of these devices is also studied, which includes a central processor module, power supplies, analog and discrete signal input/output modules, and communication modules. All these modules together offer great support in the process of the device's maintenance.

Setting up the lab starts with installing SCADA controllers and getting them ready with quick Builder in configuration Studio. The controller using Modbus gets its settings programmed, including IP addresses, scan speed, and alarm thresholds for normal and failure conditions. Thing is, after that, the experion Station is used to check if everything works right. This tool tracks the controllers status in real time, spots errors in data flows, and adjusts parameter copies to match equipment specs. That helps make the design process smoother and faster. Each step follows clearly defined instructions without extra steps or delays.

1. Tasks

1.1. Task 1. Installing SCADA controllers

To start, I opened Configuration Studio, where in the server's Control Strategy tab I navigated to "Build controllers", as shown below.

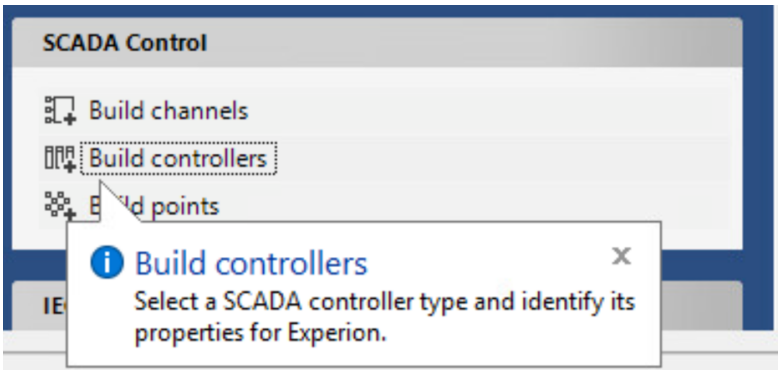

Figure 1 – Build Controllers

After that, the Quick Builder window opened, in which I clicked on the "Add item" icon, as shown below.

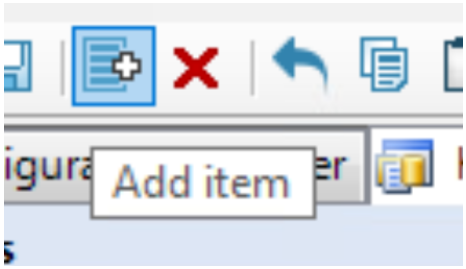

Figure 2 – “Add Item” icon

Next, in the window that opened, I selected Item Type: "Modbus Controller", and item name: "M_CR1", as indicated below.

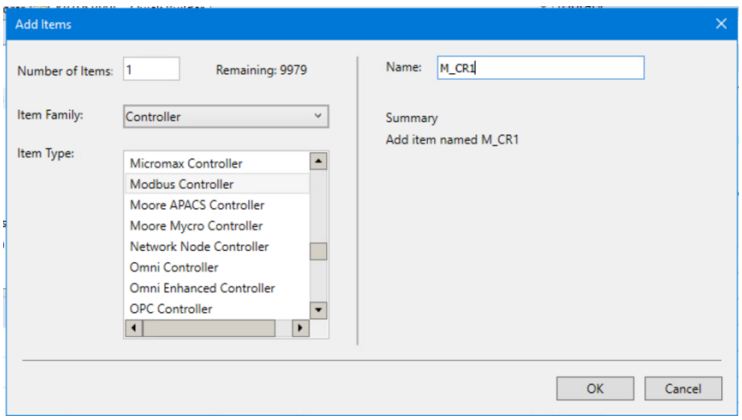

Figure 3 – “Add Item” window parameters

Before configuring the controller's IP address, you need to know the server's IP address (they are the same). To find it, I opened a command prompt window and entered the "ipconfig" command. I then found the IPv4 address: 192.168.1.6, as shown below.

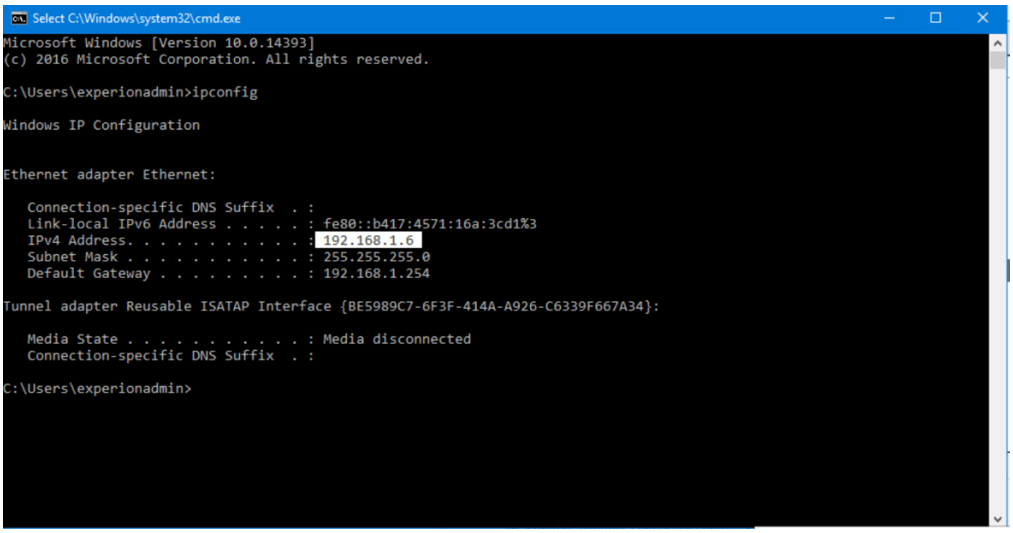

Figure 4 – Server IP configurations

Now that I knew the IP Address of the controller, I went to the Main Tab and filled in all the parameters as indicated below.

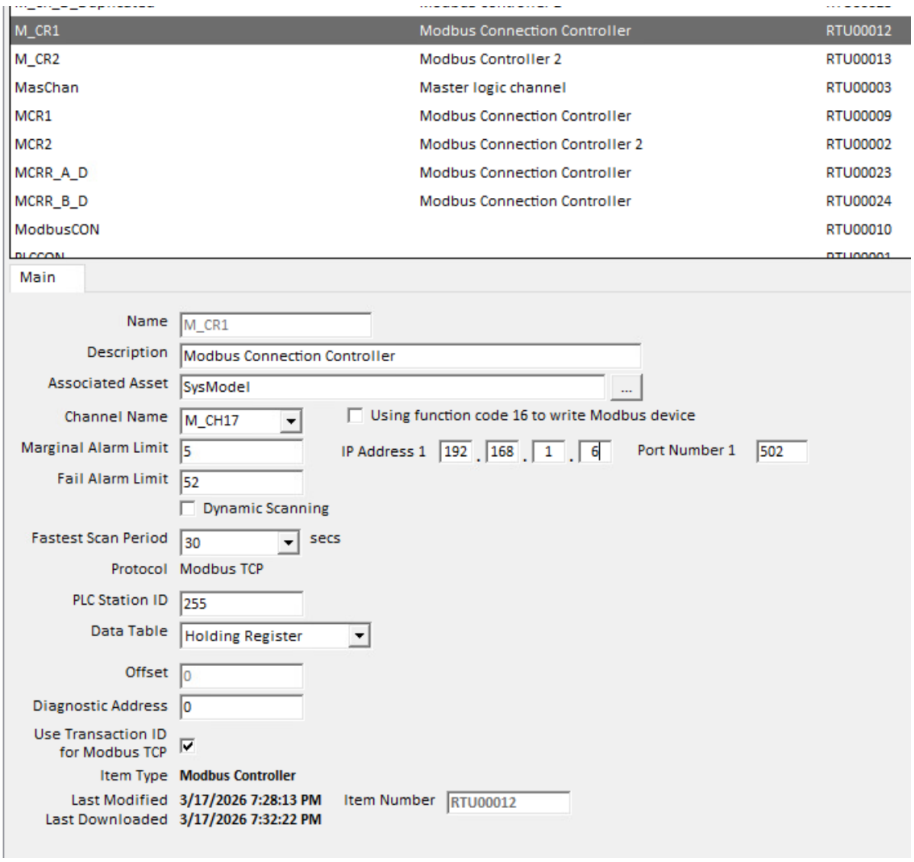

Figure 5 – M_CR1 main tab parameters

After that, I created a second controller using the same principle, with the parameters indicated below.

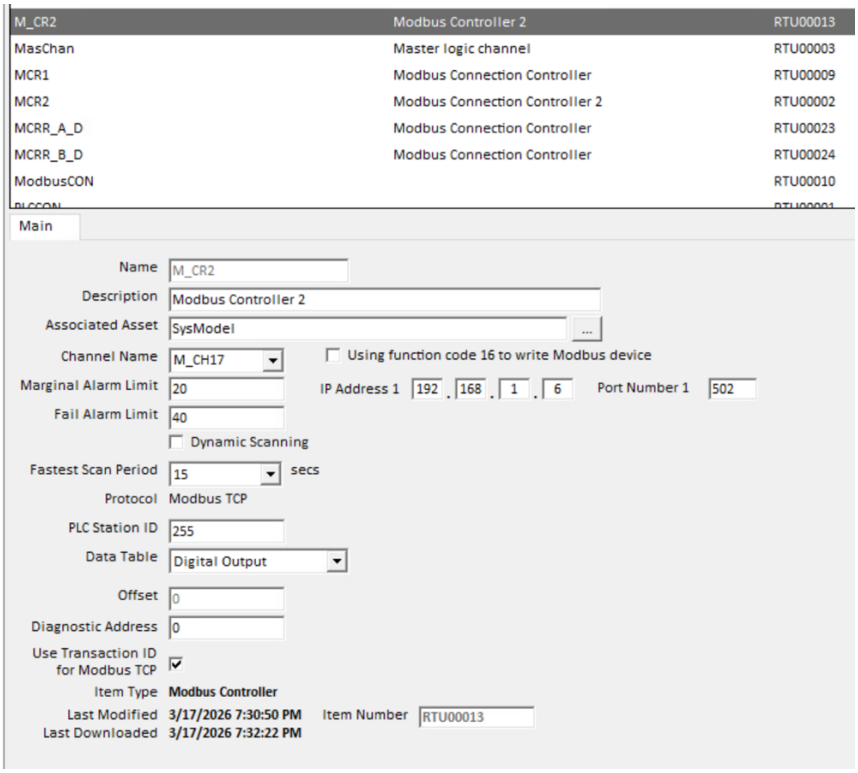

Figure 6 – M_CR2 main tab parameters

After creating two controllers, I selected them and clicked the "Download" icon. In the window that opened, I selected "Selected items only." In the "Edit options" section, I enabled "Automatically download deletions" and "Rebuild scan tables." The window is shown below.

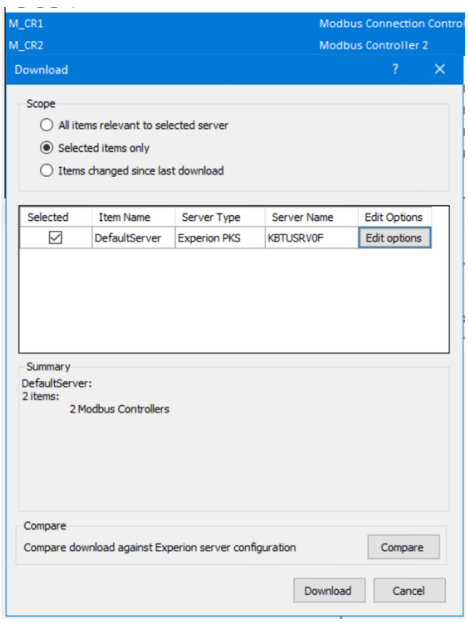

Figure 7 – Download parameters

After clicking the "Download" button, the download was successful.

1.2 Task 2. SCADA controller Status in the Station application

After that, I opened the SCADA Controllers System Status display. Path: View → System Status → SCADA Controllers, as shown below.

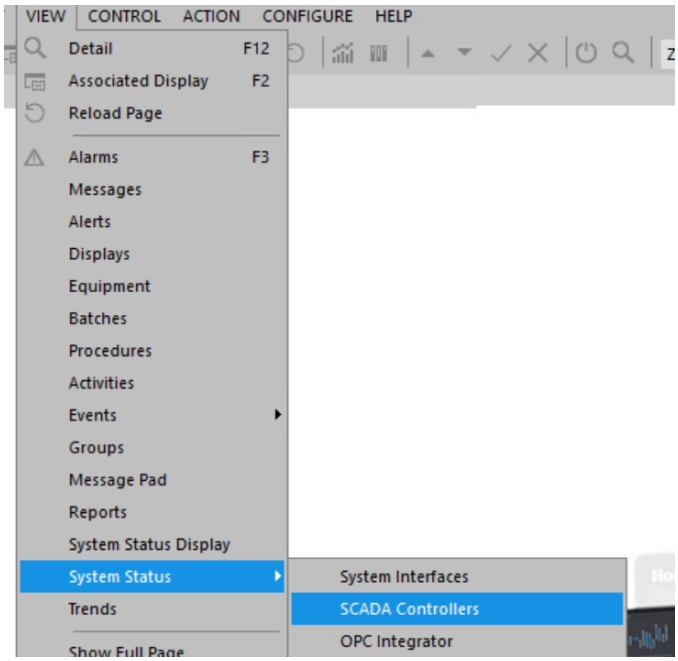

Figure 8 – SCADA Controllers

Next, in the status of the Modbus Connection Controller, I noticed the status of the M_CR1 and M_CR2 servers: Failed, as shown below.

Figure 9 – Controllers “FAILED” status

After I turned the controllers off and on, their status changed to "OK" as shown below.

Figure 10 – Controllers “OK” status

Next, I switched back to SCADA Controllers and, after expanding Channels, double-clicked M_CH17, which opened the status window. Initially, the channel status was "FAILED," but after rebooting, it changed to "OK" and the error count was 0. After a while, once the error count reached 2, the status changed to

"Marginal," indicating that when the channel reaches 2 errors, its status automatically changes to marginal. An example is shown below.

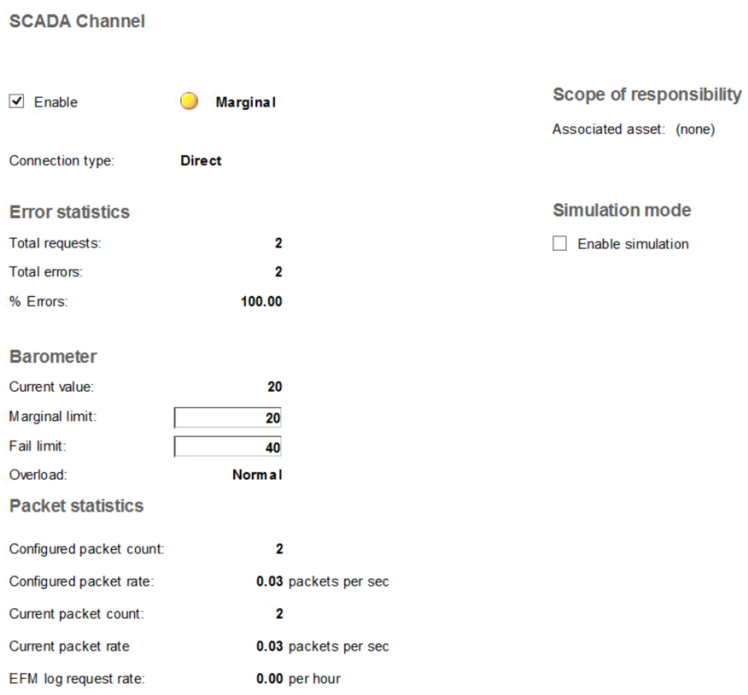

Figure 11 – Channel “Marginal” status

After some time, once the number of errors reached 4, the status changed to "Failed," indicating that when 4 errors are reached, the channel status automatically changes to "failed." An example is shown below.

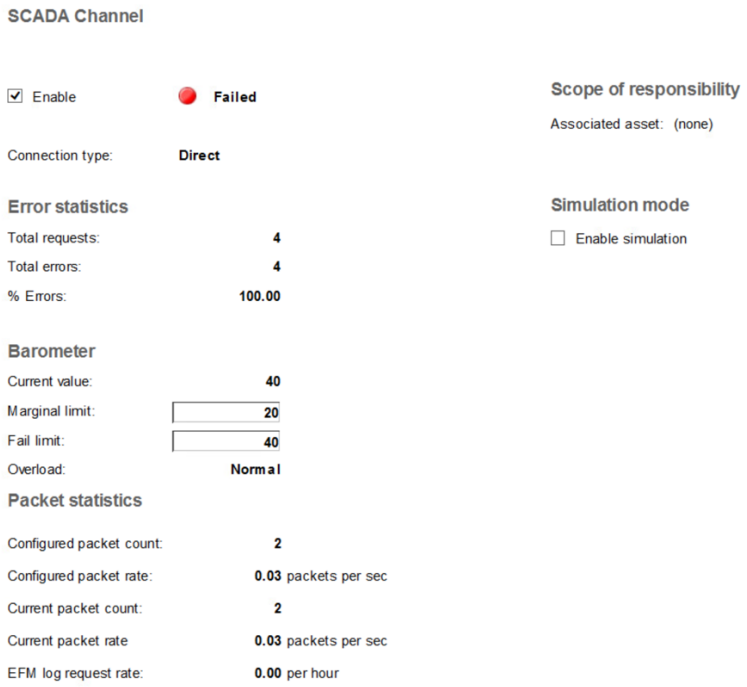

Figure 12 – Channel “Failed” status

The alarms also appeared in the System Status display in the Control field. After that, I returned to Quick Builder, clicked on M_CR1, and clicked Duplicate. Then I entered the copy parameters as shown below.

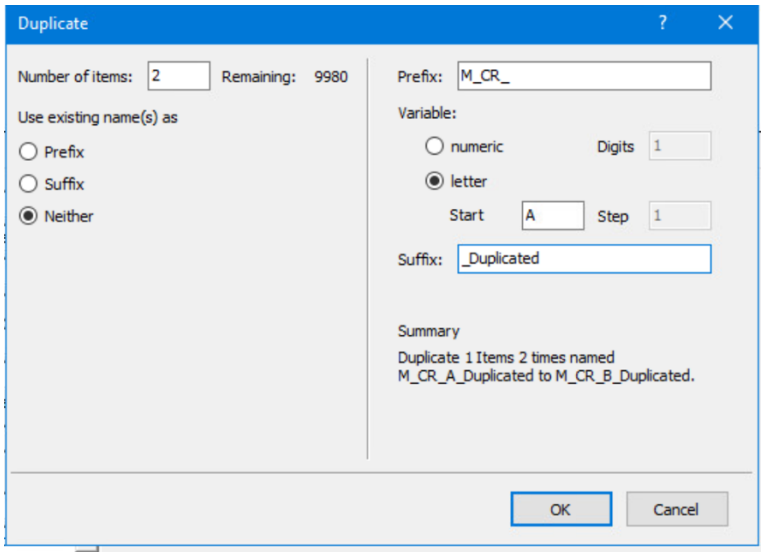

Figure 13 – Duplicate window parameters

After that I tried to upload the duplicates to the server but I got an error as below.

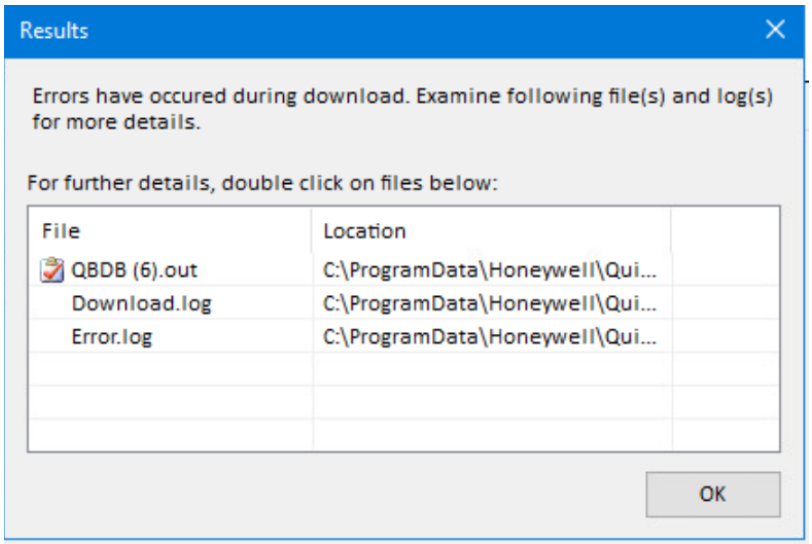

Figure 14 – Duplicate window parameters

After I opened the error details, I realized that the controller names were too long, so I shortened their names as shown below.

Figure 15 – New names for controllers

After that, I tried downloading the duplicates again. The download was successful, as described below.

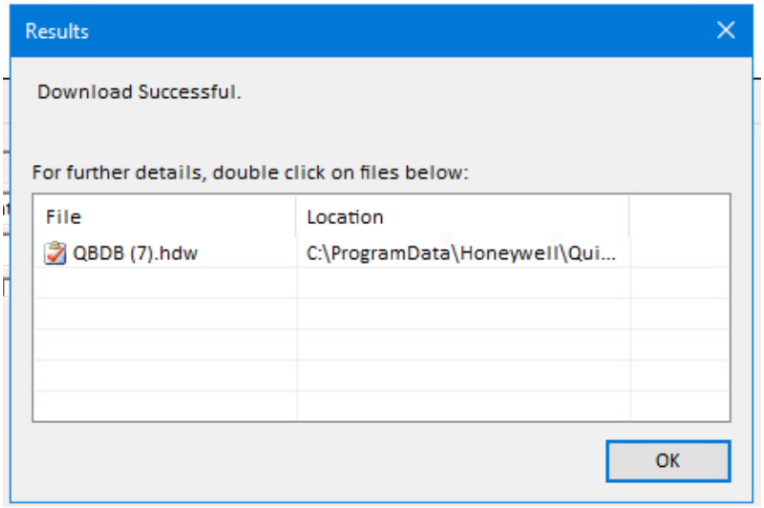

Figure 16 – Successful download

This indicates that the error has been fixed (The length of the item name should not be more than 10).

CONCLUSION

By finishing the 14th lab, the main concepts and techniques SCADA controllers were integrated into the Experion PKS AR distributed control system architecture were fully understood. Following the theoretical part, the role of PLCs was studied in depth with emphasis on their modularity of hardware and the high level of fault tolerance achieved in PLCs that enabled them to operate efficiently in harsh industrial environments.

During the practical work, apart from testing the virtual devices with the help of Quick Builder, their installation and configuration at a detailed level were also part of it. The controller which was working through Modbus protocol was made ready by setting the most important network and communication parameters. More precisely, the IP address, the Fastest Scan Period, and the system alarm limits during disconnection (Marginal and Fail Alarm Limits) were established.

WORKSHEET

Student name: Fazkullin Farid, Zhanshin Temirlan

1. (50 points) Find out information about Modbus Data Model.

Memory Block	Data type	Master Access (Read/Write)	Slave Access (Read/Write)	Description
Coils	Boolean (1 bit)	Read/Write	Read/Write	Discrete Outputs
Discrete Inputs	Boolean (1 bit)	Read only	Read/Write	Discrete Inputs
Holding Registers	Word (16 bit)	Read/Write	Read/Write	Analog Outputs
Input Registers	Word (16 bit)	Read only	Read/Write	Analog Inputs

2. (10 points) How many characters can be used in Controller`s name?

Between 1 and 10 spaceless characters

3. (20 points) Find out the information about **Data Model Addressing** of the Modbus Protocol.

In the Modbus protocol, the address space is logically segmented into four independent memory blocks corresponding to the types of data being transmitted. A prefix (the most significant digit of the register) is used to identify each block: prefix 0 defines the memory area for discrete outputs (Coils), 1 for discrete inputs (Discrete Inputs), 3 for 16-bit input registers (Input Registers), and 4 for 16-bit holding registers (Holding Registers).

4. (20 points) Find out the information about **Data Addresing Ranges** of the Modbus Protocol.

In the classic absolute address format (5-digit addressing), the Modbus protocol uses the following decimal ranges for each memory block: Coils: 00001 – 09999 ; Discrete Inputs: 10001 – 19999 ; Input Registers: 30001 – 39999 ; Holding Registers: 40001 – 49999.